

DETAILED DESCRIPTION OF THE INVENTION — SYSTEM OPERATIONAL METHODS AND USE CASE WORKFLOWS

The disclosed intraluminal physiologic monitoring device may be utilized through a variety of operational methods configured to enable reliable physiologic data acquisition across multiple usage scenarios. Operational workflows described herein are illustrative and non-limiting.

In a representative workflow, a user inserts the intraluminal monitoring device into a body lumen for the purpose of initiating a physiologic measurement session. Following insertion, the device may automatically perform calibration procedures including baseline establishment, contact verification, environmental normalization, and sensor stabilization prior to active data acquisition.

Upon successful calibration, the device may transition into a measurement state in which physiologic signals are collected continuously or periodically. Measurement duration may vary depending on user preference, predefined protocols, detected physiologic conditions, or system configuration.

During operation, the system may dynamically adapt sensing parameters in response to detected motion, posture changes, signal quality variations, or physiologic state transitions. Adaptive behavior may include modification of sampling frequency, sensor activation sequences, power consumption modes, or wireless communication timing.

In certain embodiments, the device may provide feedback to the user through companion applications, external devices, or integrated feedback mechanisms indicating calibration success, measurement readiness, or data quality status. Feedback may be visual, auditory, haptic, or algorithmically generated recommendations.

Following completion of a measurement session, collected data may undergo post-processing to generate longitudinal physiologic insights. Users may review results through abstracted indices, trend visualizations, or comparative analytics without exposure of explicit anatomical information.

Operational workflows may further include scheduled monitoring routines, guided assessment protocols, therapeutic training programs, research data collection modes, or compatibility analysis processes involving multiple synchronized devices.

Alternative workflows may support clinician-assisted operation, remote monitoring environments, automated analysis pipelines, or integration with telehealth platforms. The invention encompasses any operational method in which an intraluminal monitoring device acquires physiologic data and generates longitudinal insight through computational analysis.

All operational methods described herein are exemplary. Variations in sequence, timing, user interaction, automation level, or computational processing capable of achieving substantially similar functional outcomes are intended to fall within the scope of the disclosure.